# Development of New Methods of Adaptive Control of Non-stationary Systems During Transportation of Petroleum Products

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***Abstract*—Adaptive control systems are intelligent systems capable of changing their behavior in response to changes in the environment or in a controlled object. The key difference from traditional control systems is the ability to self-adjust and automatically adjust parameters to achieve the best performance in dynamically changing conditions. This is achieved by using special algorithms that analyze the input data, assess the state of the system and develop optimal control actions. This approach makes it possible to increase the efficiency and reliability of controlled objects, ensuring stability and accuracy even in the presence of significant disturbances and inaccuracies in the model. The development and implementation of adaptive control systems is an urgent task in various fields requiring a high degree of automation and adaptation to changing conditions.**

***Keywords—adaptive control systems, non-stationary parameters, transportation of petroleum products, processing of parametric disturbances, electromechanical systems, fuzzy logic***

## I. Introduction

The development of new methods of adaptive control is a new and relevant group facing the modern theory of automatic control and management. Resolution of the Economy of the Russian Federation of September 9, 2023 No. 1473 on the state program of the Russian Federation "Energy supply of energy efficiency".[1] This program is aimed at solving the issues of energy efficiency of all technological processes, including the processing and transportation of petroleum products. Transportation of oil and its products is a whole transport system. It includes devices and mobile means to ensure the safety of the population and industrial organizations of all types of oil cargo. At the present stage, oil refining and oil producing enterprises have new goals: tighter delivery times, low energy costs during transportation of petroleum products.

The necessary transportation conditions ensure the solution of the set goals when planning services and managing these technological processes.

When analyzing existing steam heating systems for the transportation of petroleum products and large heat losses, especially in winter, there is a need to build a new dynamic control system for these technological processes. The external environment is changing dynamically, therefore, there is a need for a system of adaptive methods and algorithms that would make it possible to move to adaptive controlled management in conditions of a changing external environment.

## II. Scientific Novelty

1. The necessary requirements for the method of mathematical description of the process of transportation of petroleum products are highlighted;

2. A description of the process of managing the transportation of petroleum products is presented;

3. A mathematical model is presented for the transportation of petroleum products by pipeline. In this model of the pumping complex used for the transportation of petroleum products by pipeline considering the resistance and pressure of the centrifugal pump based on the method of electroanalogy;

4. A mathematical model is demonstrated that takes into account the behavior of petroleum products during transportation by water transport and is described in detail in the work [2].

Adaptive automated control systems and their system description are based on the creation of new methods based on research and study of known methods. In this case, it is necessary to apply system analysis to the study of the adaptive system, determine the system goal of functioning, structure and functional content.[3]

Let's carry out the functional filling of the structure of the adaptive system. At the same time, we will formulate the requirements for mathematical description. Then we will analyze the known methods of transporting fuel oil and sulfur, as derivatives of petroleum products. The next step is to build a new method applicable to the study of adaptive systems during transportation.

An adaptive system is a new class of control associated with a qualitatively changing goal in the transportation of petroleum products, considering tight delivery times; low cost; energy savings. In this case, a mathematical description is needed that considers the specifics of this process, based on the method of system analysis, in managing transportation. We will formulate the requirements, conduct an analysis of existing methods of managing this process. Russia's transition to a market economy has created the need to implement a new management system, neither automated nor automatic, but an adaptive automatic management system for the transportation of petroleum products.

Let us formulate new methods based on known ones, after conducting a system analysis for studying adaptive systems and determine the goals, structure and content.

To fill the structure of an adaptive system, first, an analysis of existing methods is required; second, requirements for the

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mathematical description must be formulated; and third, a new adaptive system method must be constructed.

When describing a control system, we distinguish:

1. The control object (CO), which includes all processes occurring in the pumping system and in the pipeline or tanker itself;

2. We identify the disturbances:

- parametric: pressure p(t); temperature T(t); viscosity $\eta$(t); amount of heat Q(t), where t – is time.

- dynamic (structural): corrosion k(t); cavitation $k^*$(t); breakage or wear.

- emergency situations (ES), sudden increase in pressure, flow velocity, etc.

3. Processing conditions considering standard parameters—these are criteria (*K*)—creates a steady-state operating mode, eliminates unwanted effects, or completely replaces failed components. The relationship between the environment and the control system is shown in Fig. 1.

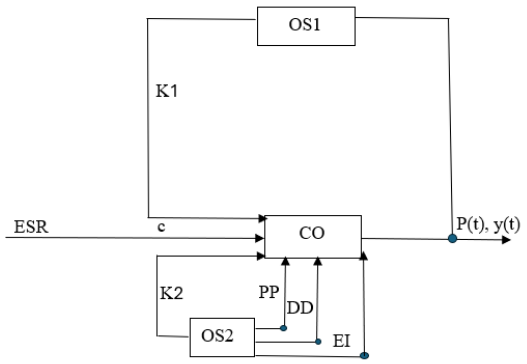


Fig. 1. Relationship between the environment and the control system: CO – control object; K – criterion, PD – parametric disturbances; DD – dynamic disturbances; OS – state processing; c – control; NRE – energy consumption standards; P – oil product transportation plan; t – time; y – oil product delivery; $\theta$ – time shift.

The control part model looks like this:

$$CO = CO(P(t), y(t), ESR(t), u(t+\theta)) = P(t+\theta), K_1(t), K_2(t) \quad (1)$$

It is possible to change these parameters if: a) create a stationary operating mode; b) eliminate undesirable effects or completely replace the failed parts.

Therefore, we will highlight the necessary requirements for the method of mathematical description of the process of transportation of petroleum products:

1. Sufficiency of the description of the transportation process in a real system, considering weather conditions, season;

2. Maintaining optimal operating modes using computer technology. Parametric and structural changes force adaptation to change in the environment, adjustment of parameters and structure of the control part during operation;

3. Description of control processes considering limitations;

4. Description of the interaction of all elements of the specified system;

5. Evaluation of the energy efficiency of the control system;

6. Accounting for changes in disturbances (parametric and dynamic) in time and space.

To create a mathematical model, it is necessary to: divide the entire transportation process into elements and describe each element separately; describe the interaction of all elements of the system.

## III. Mathematical Model

In traditional automated control systems for the transportation of petroleum products, it is impossible to take into account the optimal operating mode.

In that:

$$\sum_{j=1}^{J} a_{ij} P_j \leq b_i \quad (2)$$

Then:

$$\sum_{j=1}^{J} t_{ij} P_j \leq A_i \quad (3)$$

where $a_{ij}$ is the rate of oil product entry into the pipeline; $P_j$ is an order quantity; $b_i$ is an availability of petroleum product quantity.

Let $t_{ij}$ is mining time $j$ on equipment *i;* $t_{kr}$ is time of processing of petroleum product *r* on equipment *k*; $A_i$ is equipment operating time fund.

It is obvious that expression (3) is a constraint of (2). If there are $k$ structural elements in the system and they are therefore connected to each other, then the mathematical model will take the form:

$$y_k = S_k(M_k, x_k),$$
$$z_k = T_k(M_k, x_k)$$

where $y_k$ is the element output vector; $z_k$ is the connection vector; $T_k$ *and* $S_k$ are vector functions; $M_k$ is the control vector of the *k*-th subsystem; $x_k$ is the interaction of structural elements.

When changing nonlinear constraints, described by a vector function:

$$h_k(M_k, x_k, z_k) \geq 0$$

Interaction of elements:

$$x_k = \sum_{j=1}^{K} C_{kj} z_j \quad (4)$$

where $C_{kj}$ is the matrix of dimension.

When connected in series:

$$x_k = C_{kk1} z_{k1} \quad (5)$$

Let the objective function be:

$$F = \sum_{k=1}^{K} f_k(M_k, x_k) \rightarrow max \quad (6)$$

The goal is to determine the vectors M = {M*ₖ*}, X = {x*ₖ*}, *k* = 1, *K*, delivering the maximum of the objective function.

The solution can be conveniently implemented using the Lagrangian:

$$L = \sum_{k=1}^{K} f_k(M_k, x_k) + \sum_{k=1}^{K} \mu_k^T (T(M_k, x_k) - z_k) + \\ + \sum_{k=1}^{K} p_k^T\, (x_k - \sum_{j=1}^{K} C_{kj}, z_j\, )) + \\ + \sum_{k=1}^{K} \gamma_k^T\, h_k(M_k, x_k, z_k) \qquad (7)$$

where $\mu_k, p_k$ are the vectors of Lagrange multipliers; $\gamma_k$ is the Kuhn-Tucker multipliers.

Let the functions $T_k$, $S_k$, F be continuous and have first derivatives. Then the maximum is determined using Lagrangians [4].

In fact, the procedure for solving a single-level problem is converted into a two-level procedure.

The purpose of mathematical modeling is to study the behavior of fuel oil or sulfur during transportation, predict the results of the impact of the external environment on them, which lead to a number of disturbances, optimization of transportation control. Disturbances include weather conditions, wear, breakage, cavitation. If sulfur freezes inside the pipeline, then this object cannot be repaired.

Such disturbances are associated with changes in temperature inside the pipeline and this directly depends on the external environment, the rate of heat exchange between sulfur and the walls of the pipeline, the intensity of solar radiation and other factors.

Let us give examples of different mathematical models:

*Mathematical model for transporting petroleum products through a pipeline.*

When studying the possibility of automatic processing of disturbances in order to reduce the negative impact and to create mathematical models, it is necessary to specify methods for changing the selected parameters in each technological process.

For example, the final temperature of the petroleum product is determined using V.G. Shukhov's formula. This formula is used to calculate the temperature of the petroleum product in a pipeline during isothermal flow. It describes the temperature distribution along the entire length of the pipeline, considering heat transfer from the liquid to the environment and heat release due to internal friction forces between the layers:

$$t_k = t_0 + (t_n - t_0) \cdot e^{-mx}, \qquad (8)$$

$$mx = \frac{\lambda \cdot \rho \cdot d \cdot l}{G \cdot C_p} \qquad (9)$$

where $t_0$ is the air temperature; $t_n$ is the initial temperature of petroleum product; $mx$ is the Shukhov parameter; λ is the thermal conductivity coefficient, $W/m \cdot s$; $\rho$ is the liquid density; $d$ is the internal diameter of the pipeline; $l$ is the pipeline length; G is the liquid mass flow rate, kg/s; $C_p$ is the specific heat capacit, kJ/kg· °C.

To create a mathematical model, it is necessary to enter all the necessary parameters that participate in this process. At the same time, considering all physical processes, phase transitions [5].

The equation of conservation of mass during transportation can be written as a deformation equation:

$$\frac{\partial p}{\partial t} + \frac{\partial}{\partial x} p \cdot v = J \qquad (10)$$

where *t* is time; *x* is the coordinate; *v* is the fluid velocity; *J* is fluid flow intensity.

The equation of continuity of the jet has the form:

$$(v_1 s_1 = v_2 s_2)$$

When fuel oil is supplied through a pipeline, there is no need for receiving and draining devices, there is no water contamination of the fuel oil associated with its heating during draining, the number of service personnel is reduced, the required fuel supply is reduced, the costs of electricity and heat for own needs are reduced, and equipment wear is increased. Let us set the following parameters. Let $R_c$ be the network resistance; $P_c$ be the hydraulic network pressure; dQ be the fluid flow rate in the system. Then:

$$R_c = \frac{dP_c}{dQ}|_{Q_0} > 0 \qquad (11)$$

If $R_{c.n}$ is the resistance of a centrifugal pump; $dP_{c.n}$ is the pump head, therefore

$$R_{c.n} = \frac{dP_{c.n}}{dQ}|_{Q_0} < 0 \qquad (12)$$

Let us consider a model of a pumping complex built based on the electrical analogy method (fig. 2).

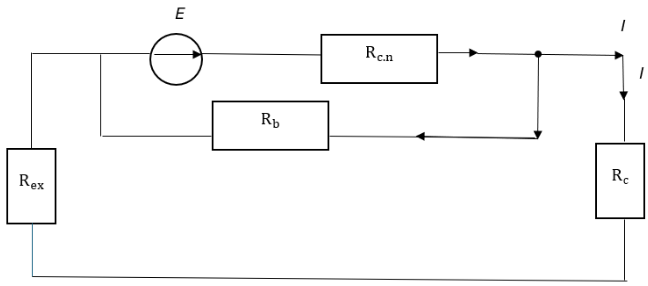


Fig. 2. Model of pumping complex.

Where $E$ is the electrical equivalent of the pressure; $R_{ex}$ is the equivalent of the resistance of the inlet pipeline; $R_b$ is the resistance of the bypass with regulation.

According to Kirchhoff's second rule for an electric circuit (complex form):

$$E(j\omega) = I(j\omega)\left[R_{c.n} + \frac{R_b \cdot (R_{ex} + R_c)}{R_b + R_{ex} + R_c}\right] \qquad (13)$$

$$R_c;\ R_{ex}$$
$$R' = R_{ex} + R_c$$

$$\frac{1}{R^n} = \frac{1}{R'} + \frac{1}{R_b} = \frac{R_b + R'}{R_b \cdot R'} = \frac{R_b + (R_c + R_{ex})}{(R_c + R_{ex}) \cdot R_b} \qquad (14)$$

$$R'' = \frac{(R_c + R_{ex}) \cdot R_b}{R_b + R_c + R_{ex}} \qquad (15)$$

$$R_{gen} = R_{c.n} + R'' = R_{c.n} + \frac{(R_c + R_{ex})}{R_b + R_c + R_{ex}} \qquad (16)$$

Voltage according to Ohm's law for a section of a circuit U=J· $R_b$

$$.\ \text{U=J} \cdot \left[R_{c.n} + \frac{(R_c + R_{\text{BX}}) \cdot R_b}{R_b + R_c + R_{\text{BX}}}\right] \qquad (17)$$

*Let's consider the differential equation for heat transfer during convection.*

Heat transfer is determined by two physical processes: conduction and transport. [6]

The solution to the differential equation of heat conduction has the form:

$$\bar{q} = \bar{q}_{tc} + \bar{q}_k = -\lambda\nabla T + \rho\bar{\omega}h \tag{18}$$

where $q$ is the heat flux density; λ is the thermal conductivity coefficient, $W/m \cdot s$; $\rho$ is the liquid density; $\omega$ is the specific enthalpy (cooling medium); $h$ is the speed (environmental parameters).

Example of heat exchange along the vertical axis:

Considering the homogeneity of the fluid, its isotropy and incompressibility, then the energy equation is:

$$\frac{\partial T}{\partial t} + W_x\frac{\partial T}{\partial x} + W_y\frac{\partial T}{\partial y} + W_z\frac{\partial T}{\partial z} = \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}\right) + \frac{q}{\rho \cdot c_p} \tag{19}$$

If:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \nabla^2 T \tag{20}$$

The energy equation is:

$$\frac{\partial T}{\partial t} = a\nabla^2 t + \frac{q_v}{\rho \cdot c_p} \tag{21}$$

In 3-dimensional space:

Along the OX axis:

$$\frac{\rho dWx}{dt} = \rho g_x - \frac{\partial \rho}{\partial x} + \mu\left(\frac{\partial^2 Wx}{\partial x^2} + \frac{\partial^2 Wy}{\partial y^2} + \frac{\partial^2 Wz}{\partial z^2}\right) \tag{22}$$

Along the OY axis:

$$\rho\frac{dWy}{dt} = \rho g_y - \frac{\partial \rho}{\partial y} + \mu\left(\frac{\partial^2 Wx}{\partial x^2} + \frac{\partial^2 Wy}{\partial y^2} + \frac{\partial^2 Wz}{\partial z^2}\right) \tag{23}$$

Along the OZ axis:

$$\rho\frac{dWz}{dt} = \rho g_z - \frac{\partial \rho}{\partial z} + \mu\left(\frac{\partial^2 Wx}{\partial x^2} + \frac{\partial^2 Wy}{\partial y^2} + \frac{\partial^2 Wz}{\partial z^2}\right) \tag{24}$$

where g is the acceleration due to gravity, $m/s^2$; $\mu$ is the viscosity coefficient.

Taking into account that the temperature coefficient of volumetric expansion is constant in a given temperature range:

$$\frac{\partial \bar{W}}{\partial t} = -\bar{g}\beta v - \frac{1}{\rho}\nabla\rho + v\nabla^2\bar{W} \tag{25}$$

where $v$ is the volume of the calculation cell.

Considering the continuity of the mass of liquid flowing out of the elementary volume and incompressibility, one can ignore the density factor, $\rho = const =>$

$$\frac{\partial Wx}{\partial x} + \frac{\partial Wy}{\partial y} + \frac{\partial Wz}{\partial z} = 0$$

The resulting differential equations for convective heat transfer describe a variety of processes, possibly taking into account geometric features, physical time, and boundary conditions [7].

When transporting petroleum products by water tanker, boundary conditions are defined such that q = 0 on the axis of symmetry and the internal walls of the tank, through which no heat is lost with the heat flux. In all other cases, geometric conditions, the presence of observers, etc., are taken into account.

Mathematical modeling allows for a wider range of study of parameters and features during the transportation of highly viscous liquids. The need to introduce assumptions and constraints limits the scope of application to the physical model we are considering.

$$\frac{\partial T}{\partial t} = a\nabla^2 T + \frac{a}{\rho c_p} \tag{26}$$

$$\frac{\partial T}{\partial t} = a\frac{\partial^2 T}{\partial x^2} + Q(t) \tag{27}$$

$$-k_1\frac{\partial T}{\partial x}x + h(T_a - T)\cdot x = Q(t) \tag{28}$$

where $a$ is the thermal conductivity coefficient of the material; $k_1$ is the thermal conductivity coefficient of walls; $h$ is the convective heat transfer coefficient; $Q(t)$ is the heat flow from the Sun.

Initial conditions:

$$T\ (x,\ 0) = T_0$$

$$0 < x < L$$

It is convenient to solve this problem using numerical methods. For simplicity, let us consider a solution using the finite difference method. Using Euler's law to approximate the derivatives, we obtain the following system of equations:

$$T_{in+1} = T_{i,n} + \Delta t \cdot [a^2 \cdot T_i + \ln 2T_{in} + T_i - \ln(\Delta x)^2] \tag{29}$$

where $T_{in}$ is the temperature value at grid node $i$ on time layer $n$; $\Delta t$ is the integration time step; $\Delta x$ is the spatial grid step.

The resulting model describes the dynamics of changes in the temperature of petroleum products in a ship's tank during transportation.

## IV. CONCLISOIN

Thus, the necessity of creating new adaptive methods of oil product transportation control is confirmed. Parametric and structural changes necessitate adaptation to changes in the external environment, adjustment of parameters and structure of the control part during operation of the automated control system.

In conclusion, we note that the developed mathematical models and the program have significant potential for use in various branches of technology and industry, where a high degree of control accuracy and reliability is required. The ability to adapt to specific operating conditions, integration with existing control and monitoring systems, as well as high efficiency indicators make the proposed approach a promising solution for the analysis and synthesis of controlled dynamic systems. [8] Further research can be aimed at optimizing computational algorithms and expanding the scope of application of the proposed methodology.

The creation and development of adaptive control systems that consider operating conditions, parameter changes and the specifics of pumped liquids are necessary conditions for increasing the efficiency and reliability of modern equipment in gas processing and other industries.

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